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SECTION : BSE – 4A

OS LAB 07

BSSE SPRING 2026

LAB 07

Task 1 Named Pipe — Send, Receive & Cleanup

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/stat.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <fcntl.h>
#include <string.h>
#define FIFO_PATH "/tmp/task1_fifo"
int main() {
    mkfifo(FIFO_PATH, 0666);
    pid_t pid = fork();
    if (pid == 0) { // CHILD (Reader)
        int fd = open(FIFO_PATH, O_RDONLY);
        char buffer[100];
        read(fd, buffer, sizeof(buffer));
        printf("Child received: %s\n", buffer);
        close(fd);
    }
    else { // PARENT (Writer)
        int fd = open(FIFO_PATH, O_WRONLY);
        char *msg = "Hello from parent through FIFO";
        write(fd, msg, strlen(msg) + 1);
        close(fd);
        wait(NULL);
        unlink(FIFO_PATH);
        printf("Parent removed FIFO and exiting.\n");
    }
    return 0;
}
```

```
shah@DESKTOP-02Q2MPC: ~  
shah@DESKTOP-02Q2MPC:~$ gcc task1.c -o task1  
shah@DESKTOP-02Q2MPC:~$ ./task1  
Child received: Hello from parent through FIFO  
Parent removed FIFO and exiting.  
shah@DESKTOP-02Q2MPC:~$
```

TASK02_ READER FILE

```
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <fcntl.h>  
#include <errno.h>  
#include <string.h>  
#include <sys/stat.h>  
#define FIFO_PATH "/tmp/task2_fifo"  
int main() {  
    mkfifo(FIFO_PATH, 0666);  
    int fd = open(FIFO_PATH, O_RDONLY | O_NONBLOCK);  
    char buf[100];  
    int retries = 0;
```

```
while (retries < 10) {

    int n = read(fd, buf, sizeof(buf));

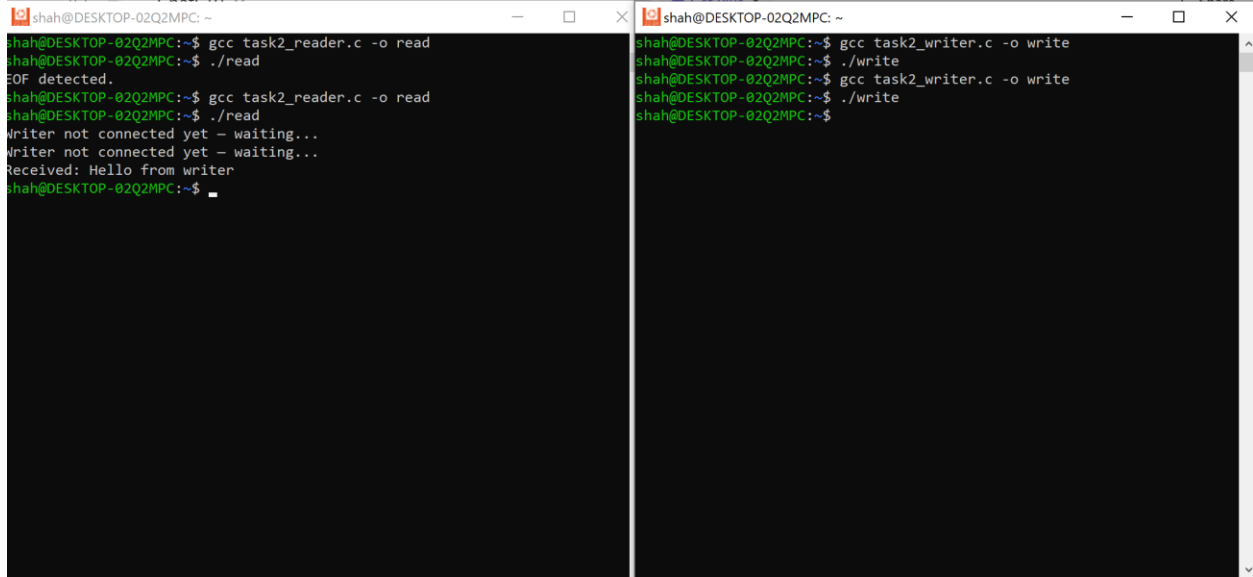
    if (n > 0) {
        printf("Received: %s\n", buf);
        break;
    }
    else if (n == -1 && errno == EAGAIN) {
        printf("No data yet — retrying...\n");
        retries++;
        sleep(1);
    }
    else if (n == 0) {
        printf("Writer not connected yet — waiting...\n");
        retries++;
        sleep(1);
    }
    }
    close(fd);
    unlink(FIFO_PATH);
    return 0;
}
```

Task_02 writer file

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <string.h>
#define FIFO_PATH "/tmp/task2_fifo"

int main() {
    int fd = open(FIFO_PATH, O_WRONLY);
    char *msg = "Hello from writer";
    write(fd, msg, strlen(msg) + 1);
    close(fd);
    return 0;
}
```

}



```
shah@DESKTOP-02Q2MPC: ~  
shah@DESKTOP-02Q2MPC:~$ gcc task2_reader.c -o read  
shah@DESKTOP-02Q2MPC:~$ ./read  
EOF detected.  
shah@DESKTOP-02Q2MPC:~$ gcc task2_reader.c -o read  
shah@DESKTOP-02Q2MPC:~$ ./read  
Writer not connected yet - waiting...  
Writer not connected yet - waiting...  
Received: Hello from writer  
shah@DESKTOP-02Q2MPC:~$  
  
shah@DESKTOP-02Q2MPC:~$ gcc task2_writer.c -o write  
shah@DESKTOP-02Q2MPC:~$ ./write  
shah@DESKTOP-02Q2MPC:~$ gcc task2_writer.c -o write  
shah@DESKTOP-02Q2MPC:~$ ./write  
shah@DESKTOP-02Q2MPC:~$
```

TASK 03

Producer1

```
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <fcntl.h>  
#include <string.h>
```

```
#define FIFO_PATH "/tmp/task3_fifo"  
#define MSGS 3
```

```
int main() {
```

```
    int fd = open(FIFO_PATH, O_WRONLY);
```

```
    for(int i = 1; i <= MSGS; i++) {
```

```
        char msg[100];
```

```
        snprintf(msg, sizeof(msg),  
        "Producer 1 (PID %d): message %d", getpid(), i);
```

```
        write(fd, msg, strlen(msg) + 1);
```

```
        sleep(1);
```

```
}

close(fd);

return 0;
}
Producer 2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <string.h>

#define FIFO_PATH "/tmp/task3_fifo"
#define MSGS 3

int main() {

    int fd = open(FIFO_PATH, O_WRONLY);

    for(int i = 1; i <= MSGS; i++) {

        char msg[100];

        snprintf(msg, sizeof(msg),
            "Producer 2 (PID %d): message %d", getpid(), i);

        write(fd, msg, strlen(msg) + 1);

        sleep(1);
    }

    close(fd);

    return 0;
}
Producer3
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
```

```
#include <fcntl.h>
#include <string.h>

#define FIFO_PATH "/tmp/task3_fifo"
#define MSGS 3

int main() {

    int fd = open(FIFO_PATH, O_WRONLY);

    for(int i = 1; i <= MSGS; i++) {

        char msg[100];

        snprintf(msg, sizeof(msg),
            "Producer 3 (PID %d): message %d", getpid(), i);

        write(fd, msg, strlen(msg) + 1);

        sleep(1);
    }

    close(fd);

    return 0;
}

Consumer
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <string.h>

#define FIFO_PATH "/tmp/task3_fifo"

int main() {

    mkfifo(FIFO_PATH, 0666);

    int fd = open(FIFO_PATH, O_RDONLY);
```

```
char buffer[200];
int n;

while((n = read(fd, buffer, sizeof(buffer))) > 0) {

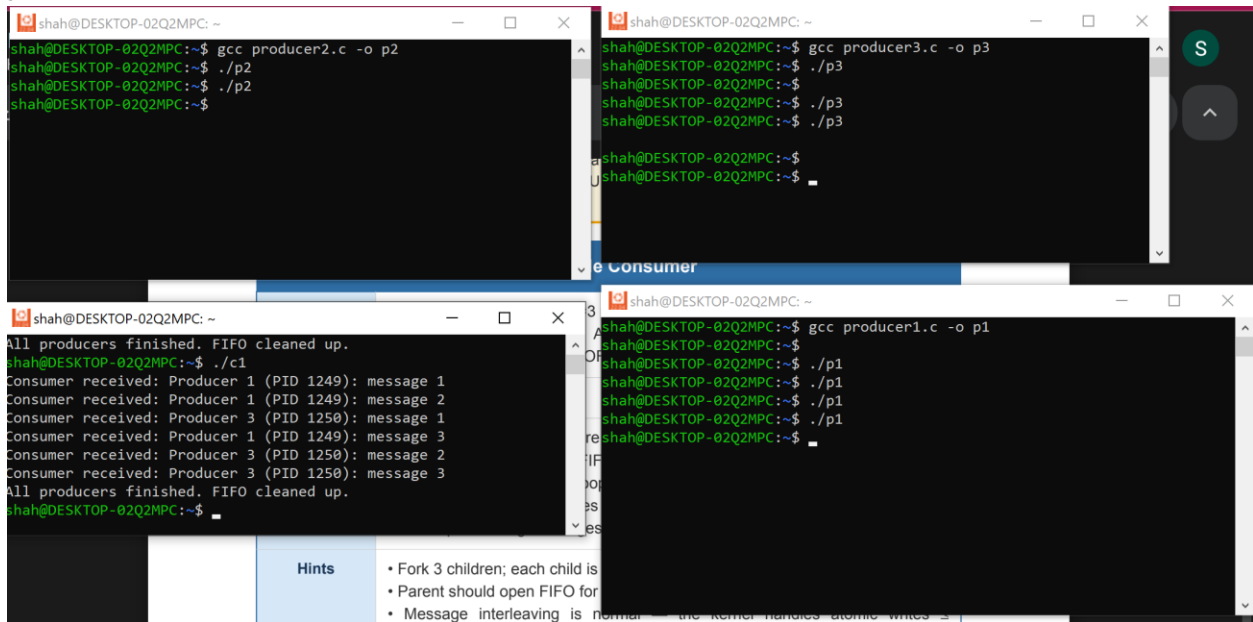
    printf("Consumer received: %s\n", buffer);
}

close(fd);

unlink(FIFO_PATH);

printf("All producers finished. FIFO cleaned up.\n");

return 0;
}
```



```
shah@DESKTOP-02Q2MPC: ~
shah@DESKTOP-02Q2MPC:~$ gcc producer2.c -o p2
shah@DESKTOP-02Q2MPC:~$ ./p2
shah@DESKTOP-02Q2MPC:~$ ./p2
shah@DESKTOP-02Q2MPC:~$

shah@DESKTOP-02Q2MPC: ~
shah@DESKTOP-02Q2MPC:~$ gcc producer3.c -o p3
shah@DESKTOP-02Q2MPC:~$ ./p3
shah@DESKTOP-02Q2MPC:~$ ./p3
shah@DESKTOP-02Q2MPC:~$ ./p3
shah@DESKTOP-02Q2MPC:~$ ./p3
shah@DESKTOP-02Q2MPC:~$

shah@DESKTOP-02Q2MPC: ~
shah@DESKTOP-02Q2MPC:~$ gcc producer1.c -o p1
shah@DESKTOP-02Q2MPC:~$ ./p1
shah@DESKTOP-02Q2MPC:~$ ./p1
shah@DESKTOP-02Q2MPC:~$ ./p1
shah@DESKTOP-02Q2MPC:~$ ./p1
shah@DESKTOP-02Q2MPC:~$ ./p1
shah@DESKTOP-02Q2MPC:~$

All producers finished. FIFO cleaned up.
shah@DESKTOP-02Q2MPC:~$ ./c1
Consumer received: Producer 1 (PID 1249): message 1
Consumer received: Producer 1 (PID 1249): message 2
Consumer received: Producer 3 (PID 1250): message 1
Consumer received: Producer 1 (PID 1249): message 3
Consumer received: Producer 3 (PID 1250): message 2
Consumer received: Producer 3 (PID 1250): message 3
All producers finished. FIFO cleaned up.
shah@DESKTOP-02Q2MPC:~$
```

Hints

- Fork 3 children; each child is
- Parent should open FIFO for
- Message interleaving is normal